

电路2复习

本次总结较匆忙，如有错误欢迎联系我。

本次主攻大是题，选择可以联系我要文档

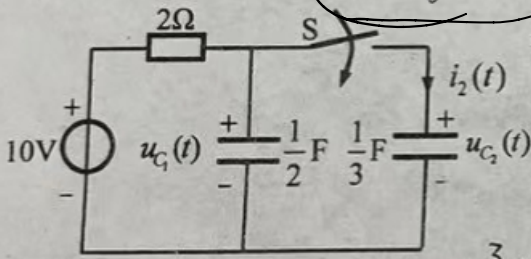
只把简单的重点总结，重难的部分并没有整理！

最后祝好运！

联系电话：187 333 13668

拉氏变换

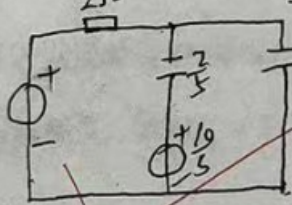
四、(15分) 图示电路中, 已知开关S 闭合前电路已处于稳态, 且 $u_{C_1}(0_-) = 0$, $t = 0$ 时开关S 闭合, 求 $t \geq 0$ 时的 $u_{C_2}(t)$ 和 $i_2(t)$ 。



$$\therefore u_{C_2}(t) = 10\varepsilon(t) - 4e^{-\frac{3}{5}t} \text{ V}$$

$$\therefore i_2(t) = C \cdot \frac{du_{C_2}(t)}{dt} = \frac{1}{3} \times (10 + \frac{12}{5}e^{-\frac{3}{5}t}) = \frac{10}{3} + \frac{4}{5}e^{-\frac{3}{5}t} \text{ A}$$

画等效电路:



$$\therefore u_{C_2}(0_-) = 0$$

$$\therefore u_{C_2}(0_+) = 0$$

$$(\frac{1}{2} + \frac{s}{2} + \frac{s}{3}) u_{C_2}(s) = \frac{10}{s} + \frac{\frac{10}{s}}{\frac{3}{5}}$$

$$u_{C_2}(s) = \frac{30 + 30s}{s(5s + 3)} = \frac{A}{s} + \frac{B}{s + \frac{3}{5}}$$

$$A = \frac{30 + 30s}{10s + 3} \Big|_{s=0} = 10, \quad B = \frac{30 + 30s}{10s + 3} \Big|_{s=-\frac{3}{5}} = -4$$

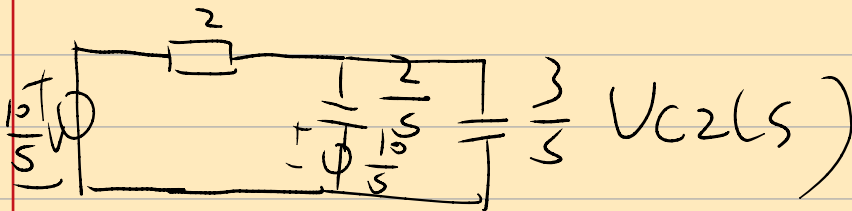
$$u_{C_2}(s) = \frac{10}{s} - \frac{4}{s + \frac{3}{5}}$$

$$\therefore u_{C_2}(t) = 10\varepsilon(t) - 4e^{-\frac{3}{5}t} \text{ V}, \quad t \geq 0$$

$$u_{C_1}(0_-) = 10 \text{ V} = u_{C_1}(0_+)$$

$$u_{C_2}(0_-) = u_{C_2}(0_+) = 0 \text{ V}$$

节点电压法



节点电压法

$$(\frac{1}{2} + \frac{s}{2} + \frac{s}{3}) U_{C_2}(s) = \frac{10}{s} + \frac{\frac{10}{s}}{\frac{3}{5}}$$

$$(\frac{3 + 5s}{6}) U_{C_2}(s) = \frac{5 + 5s}{s}$$

$$U_{C_2}(s) = \frac{5 + 5s}{s} \cdot \frac{6}{3 + 5s} = \frac{30(1 + s)}{s(3 + 5s)}$$

$$= \frac{k_1}{s} + \frac{k_2}{s + \frac{3}{5}} \quad \begin{cases} 5k_1 + k_2 = 30 \\ 3k_1 = 30 \end{cases}$$

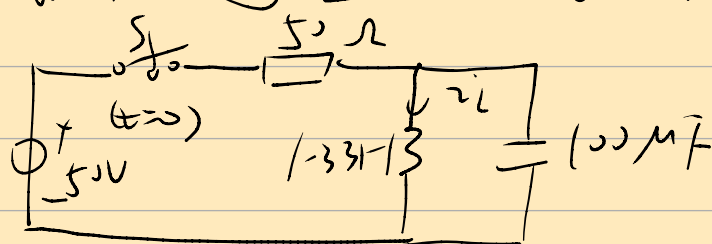
$$\therefore U_{C_2}(s) = \frac{10}{s} - \frac{4}{s + \frac{3}{5}} = \frac{10}{s} - \frac{4}{s + \frac{3}{5}}$$

$$u_{C_2}(t) = 10\varepsilon(t) - 4e^{-\frac{3}{5}t} \text{ V} \quad t \geq 0$$

$$i_{C_2}(t) = C \cdot \frac{du_{C_2}(t)}{dt} = \frac{1}{3} \cdot (10\delta(t) + \frac{12}{5}e^{-\frac{3}{5}t}) \text{ V}$$

$$= \frac{10}{3}\delta(t) + \frac{4}{5}e^{-\frac{3}{5}t} \text{ V}$$

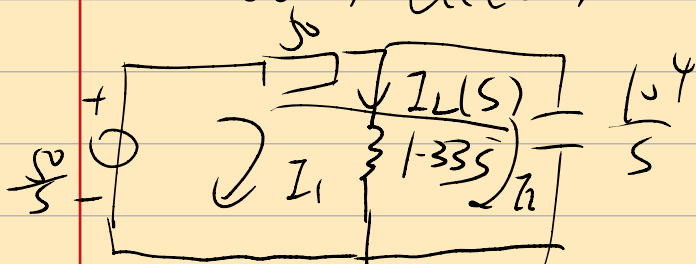
原电路处于零状态, $t=0$ 时合上开关, 求电流 i_L .



回路电流法

$$i_L(0+) = i_L(0-) = 0$$

$$u_C(0+) = u_C(0-) = 0$$



$$\begin{cases} \frac{50}{s} = (50 + 1.33s)I_1 + sI_2 \\ \frac{50}{s} = (50 + \frac{10^4}{s})I_2 + 50I_1 \end{cases}$$

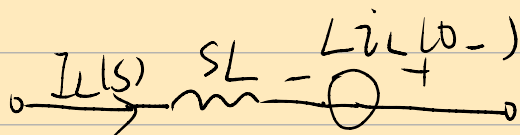
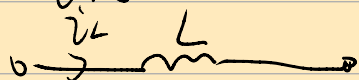
$$I_L(s) = \frac{1}{s(s+50)(s+150)} = \frac{1}{s} - \frac{1.5}{s+50} + \frac{0.5}{s+150}$$

$$i_L(t) = 1 - 1.5e^{-50t} + 0.5e^{-150t}, \quad t \geq 0.$$

拉氏变换知识点

一.	$f(t)$	$\frac{1}{s}$	t	$\frac{1}{s^2}$	ω
二.	a	$\frac{1}{s-a}$	$e^{-at} \sin \omega t$	$\frac{\omega}{(s+a)^2 + \omega^2}$	
	$\delta(t)$	1	$e^{-at} \cos \omega t$	$\frac{s+a}{(s+a)^2 + \omega^2}$	
	e^{at}	$\frac{1}{s-a}$			
	$\sin \omega t$	$\frac{\omega}{s^2 + \omega^2}$			
	$\cos \omega t$	$\frac{s}{s^2 + \omega^2}$			

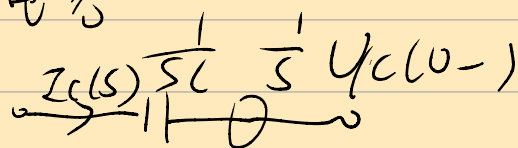
二. 电感



$$+ \quad U_L(s) \quad -$$

$$U_L(s) = sL I_L(s) - L i_L(0-)$$

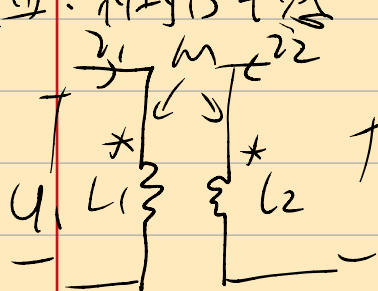
三. 电容



$$+ \quad U_C(s) \quad -$$

$$U_C(s) = \frac{1}{sC} I_C(s) + \frac{1}{s} u_C(0-)$$

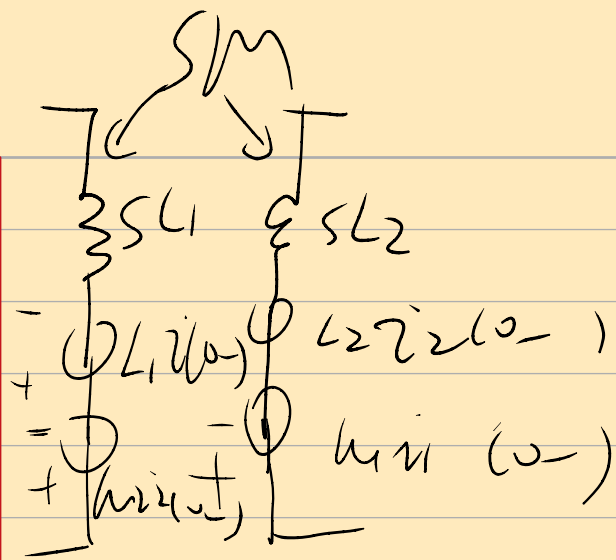
四. 耦合电感



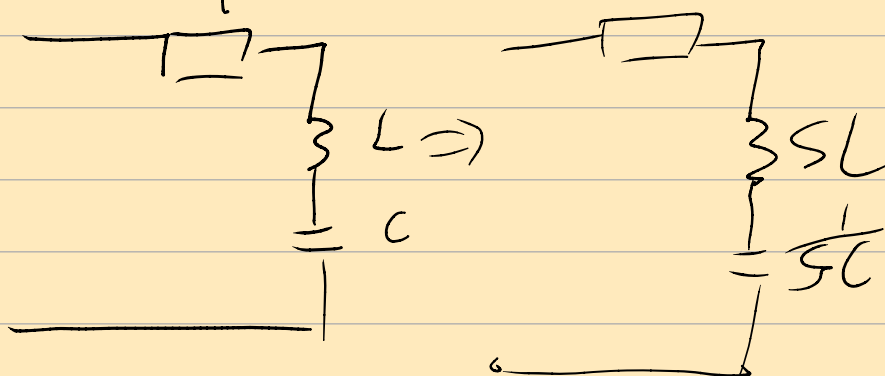
$$\begin{cases} u_1 = L_1 \frac{di_1}{dt} + M \frac{di_2}{dt} \\ u_2 = L_2 \frac{di_2}{dt} + M \frac{di_1}{dt} \end{cases}$$

$$U_1(s) = sL_1 I_1(s) - L_1 i_1(0-) + sM I_2(s) - M i_2(0-)$$

$$U_2(s) = sL_2 I_2(s) - L_2 i_2(0-) + sM I_1(s) - M i_1(0-)$$



五 RLC 串联



$$Z(s) = R + sL + \frac{1}{sC}$$

(正弦稳态: $Z = R + j\omega L + \frac{1}{j\omega C}$)

RLC 串联: 特征方程 $s^2 + \frac{R}{L}s + \frac{1}{LC} = 0$

RLC 并联: $s^2 + \frac{1}{RC}s + \frac{1}{LC} = 0$

4. 已知 RLC 并联电路的两特征根 (固有频率) 为 -1000 和 -1500, $R = 2000\Omega$, 则 $C = \underline{0.83}$

$L = \underline{0.8} \text{ H}$.

$$s^2 + \frac{1}{RC}s + \frac{1}{LC} = 0$$

$$s^2 + \frac{R}{L}s + \frac{1}{LC} = 0$$

一阶电路

2. 一 RC 放电电路, 经 0.1s 电容电压变为原来值的 20%, 时间常数为 0.062s.

3. RC 串联电路在 $t=0$ 时与 100V 电压源相接, 已知 $C=2\mu F$, 若在 $t=1s$ 时, 电容电压可由零充电到 60V, 则 R 应为 545.7kΩ.

代表两个知识点、

零输入响应 放电

$$\text{公式} \begin{cases} U_C(t) = U_0 e^{-\frac{t}{\tau}} & \tau = RC \\ I_L(t) = I_L(0) e^{-\frac{t}{\tau}} & \tau = L/R \end{cases}$$

零状态响应 充电

$$\text{公式} \begin{cases} U_C(t) = U_S (1 - e^{-\frac{t}{\tau}}) \\ I_L(t) = I_S (1 - e^{-\frac{t}{\tau}}) \end{cases}$$

τ 同上

叫时间常数

$$\begin{aligned} \text{全响应} &= U_0 e^{-\frac{t}{\tau}} + U_S (1 - e^{-\frac{t}{\tau}}) \\ &= U_S + (U_0 - U_S) e^{-\frac{t}{\tau}} \end{aligned}$$

稳态

暂态

$$1. \quad U_C(t) = U_0 e^{-\frac{t}{\tau}}$$

一般保留两位小数

$$0.2 = e^{-\frac{0.1}{\tau}}$$

$$\ln 0.2 = -\frac{0.1}{\tau}$$

$$\tau = -\frac{0.1}{\ln 0.2} = 0.062s$$

$$2. \quad U_C(t) = U_S (1 - e^{-\frac{t}{\tau}})$$

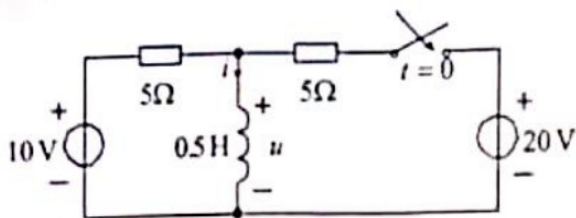
$$60 = 100 (1 - e^{-\frac{1}{R \cdot 2 \times 10^{-6}}})$$

$$0.4 = e^{-\frac{1}{2 \times 10^{-6} R}}$$

$$\ln 0.4 = -\frac{1}{2 \times 10^{-6} R}$$

$$\begin{aligned} R &= \frac{1}{\ln 0.4 \cdot 2 \times 10^{-6}} \\ &= 545.7k\Omega \end{aligned}$$

三、(15分) 图示电路原已处于稳态, 当 $t=0$ 时开关闭合, 求 $i(t)$, $u(t)$, $t \geq 0$ 。



$$\text{求: } u(0^-) = i_L(0^-) = \frac{10V}{5\Omega} = 2A$$

$$i_L(0^+) = i_L(0^-)$$

$$\text{当开关闭合后 } i_L(\infty) = \frac{10V}{5\Omega} + \frac{20V}{5\Omega} = 6A$$

$$\text{根据三要素法 } i(t) = i_L(\infty) + (i_L(0^+) - i_L(\infty))e^{-\frac{t}{\tau}}$$

$$\therefore i(t) = 6 - 4e^{-\frac{t}{\tau}}$$

$$\therefore \text{可求得 } i(t) = 6 - 4e^{-\frac{t}{\tau}} (A)$$

$$u(t) = \frac{d}{dt} \int i(t) dt$$

先求 $u(0^-)$ 、 $i(0^-)$ 再求 τ 的，
换路定则之类的。

$$i(0^+) = i_L(0^-) = \frac{10}{5} = 2A$$

$$R_{eq} = \text{等效电阻}$$

$$\tau = \frac{0.5}{2.5} = \frac{1}{5}$$

$$i_L(\infty) = \frac{10}{5} + \frac{20}{5} = 6A$$

$$i_L(t) = i_L(\infty) + [i_L(0^+) - i_L(\infty)]e^{-\frac{t}{\tau}}$$

$$= 6 + (2 - 6)e^{-5t} = 6 - 4e^{-5t}$$

$$u_L(t) = L \frac{di_L(t)}{dt} = 0.5 \times [-4 \times (-5)] e^{-5t} = 10e^{-5t}$$

全响应用三要素法

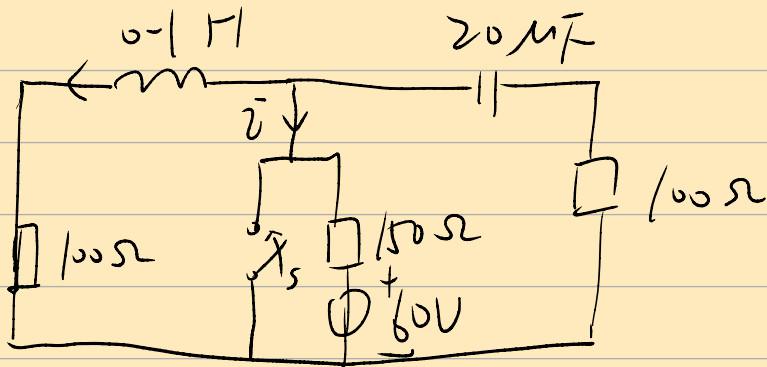
$$u(t) = u(\infty) + [u(0^+) - u(\infty)]e^{-\frac{t}{\tau}}$$

$$i(t) = i(\infty) + [i(0^+) - i(\infty)]e^{-\frac{t}{\tau}}$$

$$\tau = RC \text{ 或 } L/R$$

$$R_{eq} = \text{等效电阻}$$

∞ 时 0^+ 时 和 R 。

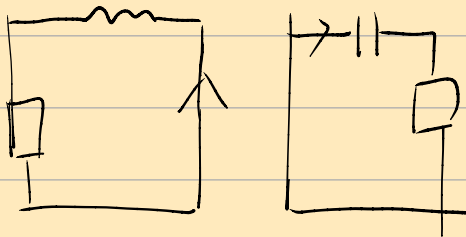


$t=0$ 时开关闭合 求电流 i

假设为二阶，可以折成二个一阶，直接套公式
不用拉氏变换

$$i_L(0+) = i_L(0-) = \frac{60}{\frac{100}{20} + 150} = 0.24 \text{ A}$$

$$U_C(0+) = U_C(0-) = \frac{100}{250} \cdot 60 = 24 \text{ V}$$



$$i = i_1 + i_2$$

两个零输入 (放电) $\tau = L/R = 1 \times 10^{-3} \text{ s}$

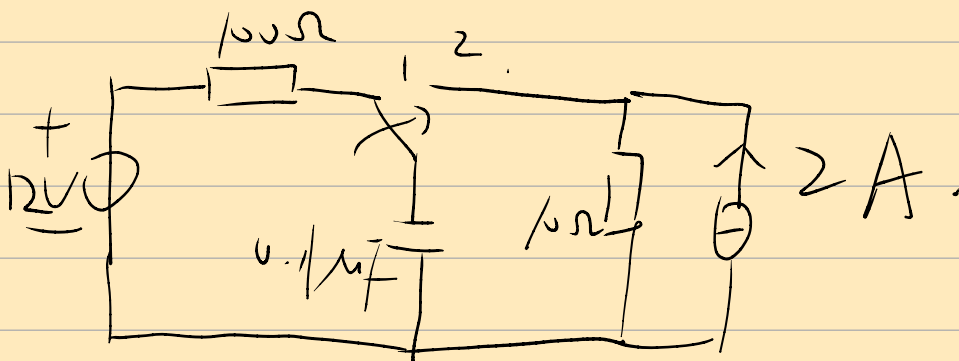
$$i_L(t) = i_L(0+) e^{-\frac{t}{\tau}} = 0.24 e^{-10^3 t}$$

$$U_C(t) = U_C(0+) e^{-\frac{t}{\tau}} = 24 e^{-500 t} \quad \tau = RC = 2 \times 10^{-3} \text{ s}$$

$$i_C(t) = C \cdot \frac{dU_C(t)}{dt} = 20 \times 10^{-6} \times 24 \times (-500) e^{-500 t}$$

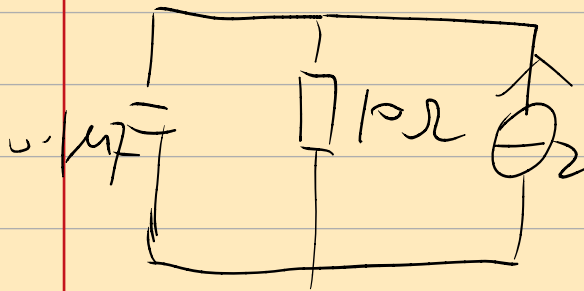
$$i = -i_C(t) - i_L(t) = 0.24 (e^{-500 t} - e^{-1000 t}) \text{ A}$$

$$= 0.24 (e^{-500 t} - e^{-1000 t}) \text{ A}$$



求 $U_C(t)$

$$u_C(0+) = u_C(0-) = U_S = 12V$$



$$\frac{1}{\tau} = \frac{1}{RC}$$

$$\tau = \frac{10 \times 10^{-6}}{10^{-4}} = 10^{-4}$$

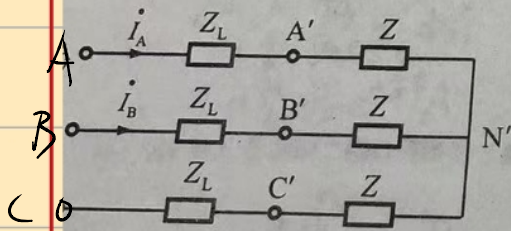
$$u_C(\infty) = 20V$$

$$u_C(0+) = 12V$$

$$\begin{aligned} \therefore u_C(t) &= u_C(\infty) + [u_C(0+) - u_C(\infty)] e^{-\frac{t}{\tau}} \\ &= 20 - 8e^{-10^4 t} \end{aligned}$$

三相电路

二、(15分) 图示对称三相电路中，已知负载端线电压 $\dot{U}_{AB'} = 1143.16 \angle 0^\circ \text{ V}$ ， $Z = 30 \angle 60^\circ \Omega$ ， $Z_L = 2.236 \angle 63.435^\circ \Omega$ ，求 $\dot{I}_A$ 及 $\dot{U}_{AB}$ 。



解: $\dot{U}_{AN'} = \frac{1143.16 \angle 30^\circ}{\sqrt{3}} = 660 \angle 30^\circ \text{ V}$

$\dot{I}_A = \frac{660 \angle 30^\circ}{30 \angle 60^\circ} = 22 \angle -30^\circ \text{ A}$

$\dot{U}_{AN'} = \dot{I}_A (Z_L + Z)$
 $= 22 \angle -30^\circ \times (16 + j27.98) = 709.1 \angle -27.16^\circ$

$\dot{U}_{AB} = \sqrt{3} \dot{U}_{AN'} \angle 30^\circ$
 $= 1228.19 \angle 0.24^\circ \text{ V}$

$\therefore \dot{I}_A = 22 \angle -30^\circ \text{ A}$

$\dot{U}_{AB} = 1228.19 \angle 0.24^\circ \text{ V}$

$\dot{U}_{A'B'} = 1143.16 \angle 0^\circ \text{ V}$

$\therefore \dot{U}_{A'N'} = \frac{1}{\sqrt{3}} \cdot 1143.16 \angle 30^\circ \text{ V} = 660 \angle 30^\circ \text{ V}$

$\dot{I}_A = \frac{\dot{U}_{A'N'}}{Z} = \frac{660 \angle 30^\circ}{30 \angle 60^\circ} = 22 \angle -30^\circ \text{ V}$

$\dot{I}_{AN'} = \dot{I}_A$

$\dot{U}_{AN'} = \dot{I}_{AN'} \cdot (Z_L + Z) = 22 \angle -30^\circ \cdot (Z_L + Z)$

求: $Z_L + Z = 30 \angle 60^\circ + 2.236 \angle 63.435^\circ = 15 + j25.98 + 1 + j2$
 $= 16 + j27.98 = 32.23 \angle 60.24^\circ$

$\therefore \dot{U}_{AN'} = 22 \angle -30^\circ \cdot 32.23 \angle 60.24^\circ$
 $= 709.1 \angle -29.76^\circ$

$\dot{U}_{AB} = \sqrt{3} \dot{U}_{AN'} \angle 30^\circ = 1228.2 \angle 0.24^\circ \text{ V}$

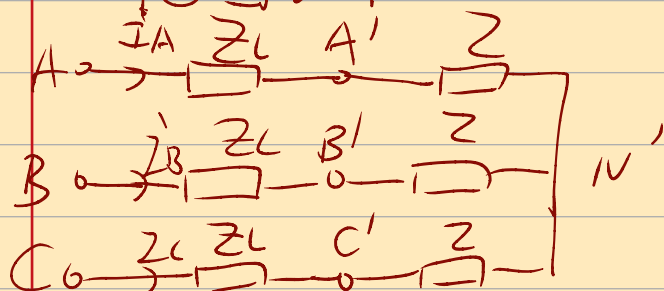
知识点:

例题:

1. 计算器: $15 + j30 \rightarrow 30 \angle 30^\circ$ 按 $\rightarrow$
 $\text{Pol}(30, 30) \rightarrow 30 \angle 30^\circ$ 按 $\rightarrow$
 $\text{Pol}(30, 30) \rightarrow 30 + j25.98$

1. $30\angle 30^\circ \rightarrow 30\angle 120^\circ$
 SHIFJ $30\angle 30^\circ \rightarrow 30\angle 120^\circ$

2. 把图记住



3. $\Delta \rightarrow Y$ $Z_Y = \frac{1}{3} Z_\Delta$ ✗
 都用 Y 形来做题.

Y: $U_L = \sqrt{3} U_p \angle 30^\circ$ $I_L = I_p$

$\Delta: I_L = \sqrt{3} I_p \angle 30^\circ$ $U_L = U_p$

5. 如果对称, 用归一相法.

各路之间差 120° , 记作不要超过 180° .

6. $P = 3 U_p I_p \cos \varphi$ 变形自行查看 (可以不要).

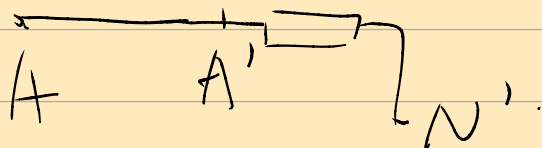
3. 主观题 (15分)

对称三相电路的线电压为 $U_L = 380V$ 负载阻
 抗 $Z = 12 + 16j(\Omega)$

无线路阻抗, 试求: 当负载星形联结时的线电流有效值及吸收总的有功功率 P

请输入内容 - $U_p = 220V$

解:



$$U_L = 380V$$

$$\dot{U}_P = 220 \angle 0^\circ V$$

$$I_L = I_P = \frac{U_P}{Z} = \frac{220 \angle 0^\circ}{20 \angle 53.1^\circ} = 11 \angle -53.1^\circ$$

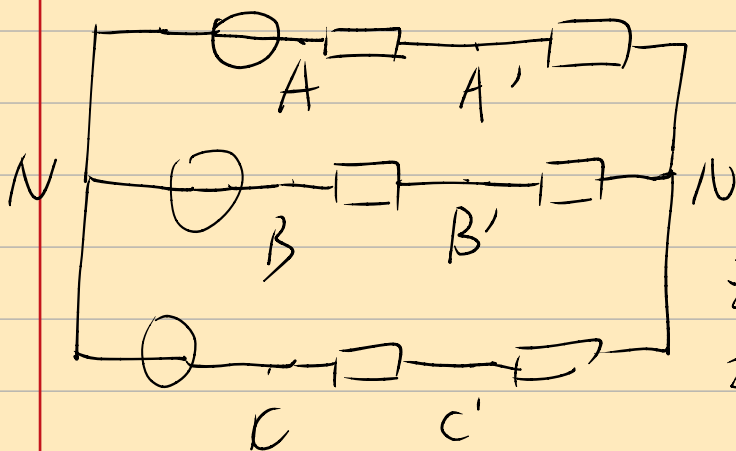
有效值 $I_L = 11A$

$$P = 3 U_P I_P \cos \varphi = 3 \times 220 \times 11 \cdot \cos 53.1^\circ = 4359W$$

三相注意计算 (画图略)

11-2 已知对称三相电路的线电压 $U_l = 380V$ (电源端), 三角负载阻抗 $Z = (4.5 + j14)\Omega$, 端线阻抗 $Z_l = (1.5 + j2)\Omega$ 。求线电流和负载的相电流, 并作相量图。

本题是 Δ 连接 $I_P = \sqrt{3} I_L \angle 30^\circ$



$$Z_Y = \frac{1}{3} Z = 1.5 + j4.67 \Omega$$

$$\dot{U}_{AN'} = 220 \angle 0^\circ$$

$$\dot{I}_A = \frac{\dot{U}_{AN'}}{Z + Z_Y} = 30.08 \angle -65.78^\circ$$

$$\dot{I}_B = 30.08 \angle -155.78^\circ$$

$$\dot{I}_C = 30.08 \angle 54.22^\circ A$$

$$\dot{I}_{A'B'} = \frac{\dot{I}_A}{\sqrt{3}} \angle 30^\circ = 17.37 \angle -35.78^\circ A$$

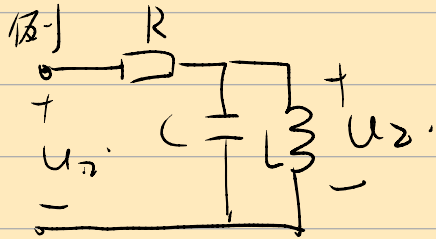
$$\dot{I}_{B'C'} = 17.37 \angle -125.78^\circ A$$

$$\dot{I}_{C'A'} = 17.37 \angle 84.22^\circ A$$

网络函数

用会拉氏正逆变换就行，没有特别重点。（卷积不看了）

$$H(s) = \frac{R(s) \rightarrow \text{零状态}}{E(s) \rightarrow \text{激励}}$$

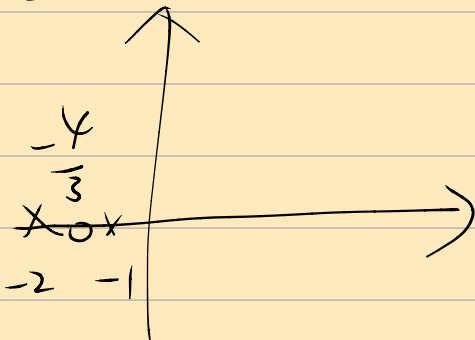


$$\frac{U_2(s)}{U_1(s)} = \frac{2s}{s^2 + 5s + 6} \quad \text{选定 } R = 1k\Omega, \text{ 则 } L = \quad C =$$

某线性电路的冲激响应 $h(t) = e^{-t} + 2e^{-2t}$
试求网络函数，并画出零、极点图

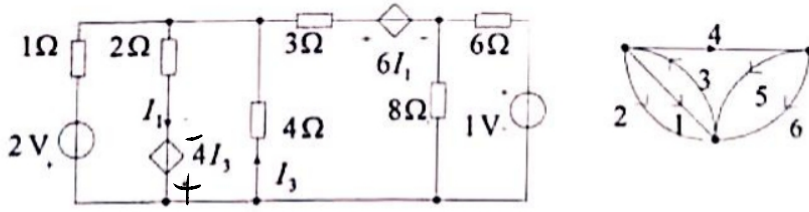
$$H(s) = \frac{1}{s+1} + \frac{2}{s+2} = \frac{s+2+2s+2}{(s+1)(s+2)} = \frac{3s+4}{(s+1)(s+2)}$$

$$p_1 = -1 \quad p_2 = -2 \quad z_1 = -\frac{4}{3}$$

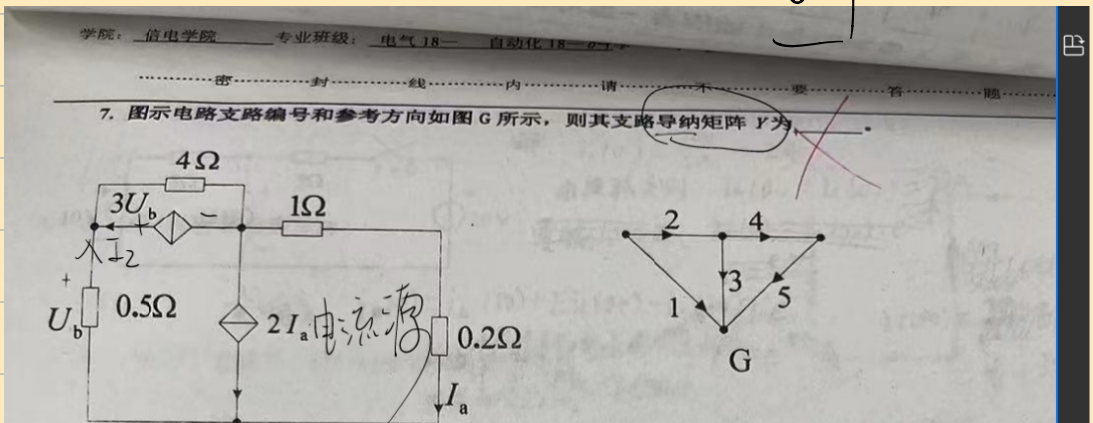


矩阵

6. 图 G 中各支路已编号及定向, 电路的支路阻抗矩阵 Z

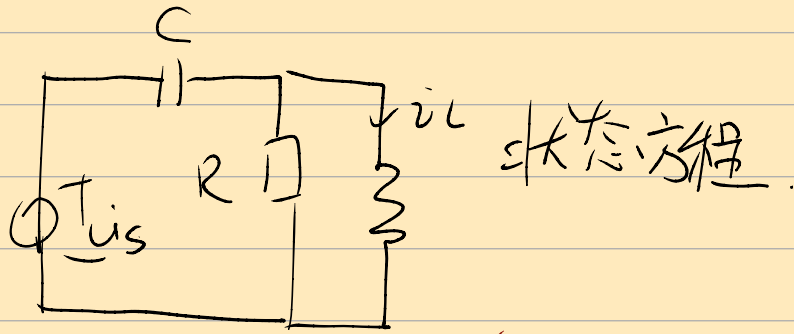


$$Z = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 8 & 0 \\ 0 & 0 & 0 & 0 & 0 & 6 \end{bmatrix}$$



$$Y = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1/4 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 5 \end{bmatrix}$$

基础部分在此不列出了,有需要联系我.



思路: 简单的用直观法. 对 C 用 KCL .
L 用 KVL .

等式一边为导数, 一边为状态量 + 激励力的形式.

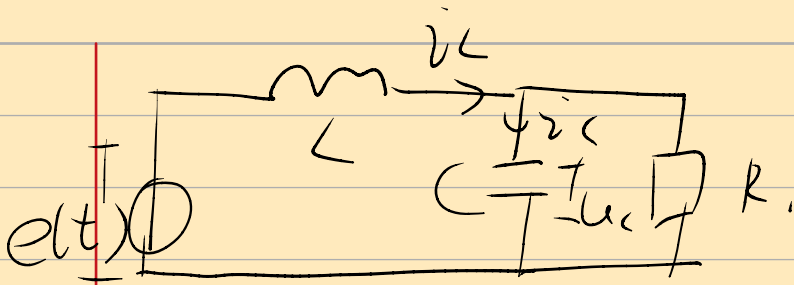
$$C \cdot \frac{du_C}{dt} = i_R + i_L = \frac{u_s - u_C}{R} + i_L$$

$$L \cdot \frac{di_L}{dt} = u_s - u_C$$

$$\frac{du_C}{dt} = \frac{-1}{CR} u_C + \frac{1}{C} i_L + \frac{1}{CR} u_s$$

$$\frac{di_L}{dt} = \frac{-1}{L} u_C + \frac{1}{L} u_s$$

$$\therefore \begin{bmatrix} u_C' \\ i_L' \end{bmatrix} = \begin{bmatrix} \frac{-1}{CR} & \frac{1}{C} \\ \frac{-1}{L} & 0 \end{bmatrix} \begin{bmatrix} u_C \\ i_L \end{bmatrix} + \begin{bmatrix} \frac{1}{CR} \\ \frac{1}{L} \end{bmatrix} u_s$$



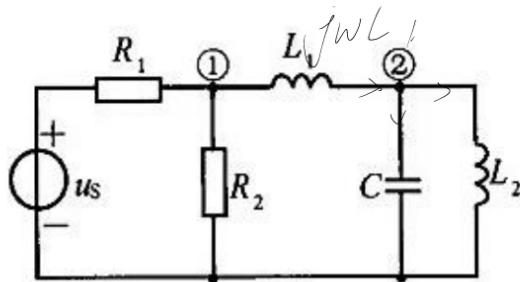
$$C \frac{du_C}{dt} = i_L - \frac{u_C}{R}$$

$$L \frac{di_L}{dt} = e(t) - u_C$$

$$\begin{bmatrix} u_C' \\ i_L' \end{bmatrix} = \begin{bmatrix} -\frac{1}{RC} & \frac{1}{C} \\ \frac{1}{L} & 0 \end{bmatrix} \begin{bmatrix} u_C \\ i_L \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{L} \end{bmatrix} e(t)$$

6.

7 列出题 15—17 图所示电路的状态方程。若选结点 ① 和 ② 的结点电压为输出量, 写出输出方程。



$$\frac{du_C}{dt} = \frac{1}{C} i_{L1} - \frac{1}{C} i_{L2}$$

$$\frac{di_{L1}}{dt} = i_{L1} - i_{L2}$$

$$i_{L2} = \frac{u_C}{R_2}$$

较难(可以暂时不看)

$$C \frac{du_C}{dt} = i_{L1} - i_{L2}$$

$$u_{R2} = R_2 (i_{L1} - i_{L2})$$

$$i_{L1} = \frac{u_S - u_{R2}}{R_1}$$

$$L_1 \frac{di_{L1}}{dt} = u_{R2} - u_C$$

$$u_{R2} = -\frac{R_1 R_2}{R_1 + R_2} \frac{di_{L1}}{dt} + \frac{R_2}{R_1 + R_2} u_S$$

$$L_2 \frac{di_{L2}}{dt} = u_C$$

$$\frac{du_c}{dt} = -\frac{1}{C} i_{L1} - \tau' i_{L2}$$

$$\frac{di_{L1}}{dt} = -\frac{1}{L_1} u_c - \frac{R_1 R_2}{L_1 (R_1 + R_2)} i_{L1} + \frac{R_2}{L_1 (R_1 + R_2)} u_s$$

$$\frac{di_{L2}}{dt} = \frac{1}{L_2} u_c$$

$$\begin{bmatrix} \frac{du_c}{dt} \\ \frac{di_{L1}}{dt} \\ \frac{di_{L2}}{dt} \end{bmatrix} = \begin{bmatrix} 0 & \frac{1}{C} & -\tau' \\ -\frac{1}{L_1} & -\frac{R_1 R_2}{L_1 (R_1 + R_2)} & 0 \\ \frac{1}{L_2} & 0 & 0 \end{bmatrix} \begin{bmatrix} u_c \\ i_{L1} \\ i_{L2} \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{R_2}{L_1 (R_1 + R_2)} \\ 0 \end{bmatrix} u_s(t)$$

$$u_{s1} = u_{R2} = \frac{R_1 R_2}{R_1 + R_2} i_{L1} + \frac{R_2}{R_1 + R_2} u_s$$

$$u_{n2} = u_c$$

$$\begin{bmatrix} u_{s1} \\ u_{n2} \end{bmatrix} = \begin{bmatrix} 0 & \frac{R_1 R_2}{R_1 + R_2} & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} u_c \\ i_{L1} \\ i_{L2} \end{bmatrix} + \begin{bmatrix} \frac{R_2}{R_1 + R_2} \\ 0 \end{bmatrix} u_s(t)$$

二端口网络

1. 4个参数

$$\sum \begin{bmatrix} \dot{U}_1 \\ \dot{U}_2 \end{bmatrix} = \sum \begin{bmatrix} \dot{I}_1 \\ \dot{I}_2 \end{bmatrix}$$

$$\Upsilon \begin{bmatrix} \dot{I}_1 \\ \dot{I}_2 \end{bmatrix} = \Upsilon \begin{bmatrix} \dot{U}_1 \\ \dot{U}_2 \end{bmatrix}$$

$$\Upsilon \begin{bmatrix} \dot{U}_1 \\ \dot{I}_1 \end{bmatrix} = \Upsilon \begin{bmatrix} \dot{U}_2 \\ -\dot{I}_2 \end{bmatrix}$$

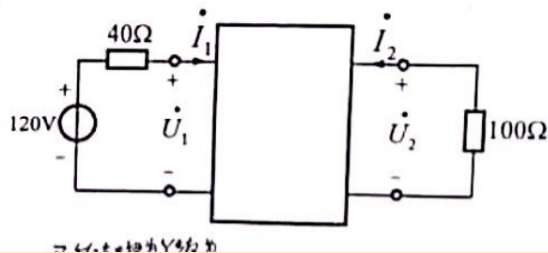
$$H \begin{bmatrix} \dot{U}_1 \\ \dot{I}_2 \end{bmatrix} = H \begin{bmatrix} \dot{I}_1 \\ \dot{U}_2 \end{bmatrix}$$

五、(15分) 若图示二端口网络的Z参数矩阵为

$$Z = \begin{bmatrix} 50 & 10 \\ 30 & 20 \end{bmatrix} \Omega$$

试计算100Ω电阻消耗的功率。

$$\begin{aligned} \text{由Z参数方程得 } \dot{U}_1 &= Z_{11}\dot{I}_1 + Z_{12}\dot{I}_2 \\ \dot{U}_2 &= Z_{21}\dot{I}_1 + Z_{22}\dot{I}_2 \end{aligned}$$



$$\dot{U}_1 = 50\dot{I}_1 + 10\dot{I}_2$$

$$\dot{U}_2 = 30\dot{I}_1 + 20\dot{I}_2$$

$$\begin{cases} \dot{U}_1 = 120V - 40\dot{I}_1 \\ \dot{U}_2 = 100\dot{I}_2 \end{cases}$$

联立解得

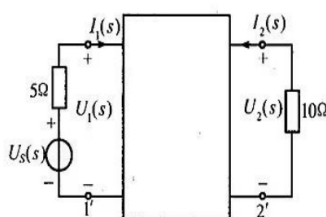
$$\dot{I}_2 = -\frac{12}{55} A$$

$$P = I_2^2 \times 100 = 11.76 W$$

电路如题16-9图所示, 已知二端口的H参数矩阵为

$$H = \begin{bmatrix} 40 & 0.4 \\ 10 & 0.1 \end{bmatrix}$$

求电压转移函数 $U_2(s)/U_s(s)$ 。



题16-9图

$$U_1(s) = 40I_1(s) + 0.4U_2(s)$$

$$45I_1(s) + 0.4U_2(s) = U_s(s)$$

$$\dot{U}_1 = 40\dot{I}_1(s) + 0.4\dot{U}_2(s)$$

$$\dot{U}_2 = 10\dot{I}_1(s) + 0.1\dot{U}_2(s)$$

$$\dot{U}_1(s) = U_s(s) - 5\dot{I}_1(s)$$

$$\dot{U}_2(s) = -10\dot{I}_2(s)$$

$$\dot{U}_2(s)$$

$$\rightarrow \frac{U_s(s)}{\dot{U}_2(s)} = \frac{1}{0.5} = -2$$

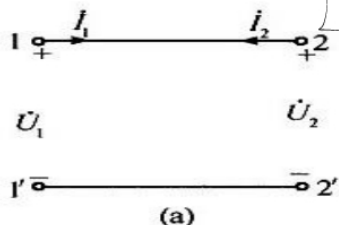
电压转移函数。

求 T 参数矩阵

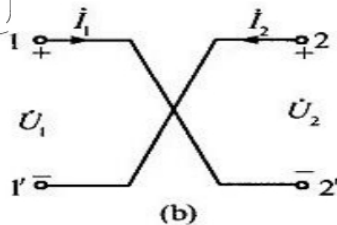
$$\begin{bmatrix} U_1 \\ I_1 \end{bmatrix} = T \begin{bmatrix} U_2 \\ -I_2 \end{bmatrix}$$

$$\begin{cases} \dot{U}_1 = -\dot{U}_2 \\ \dot{I}_1 = \dot{I}_2 \end{cases}$$

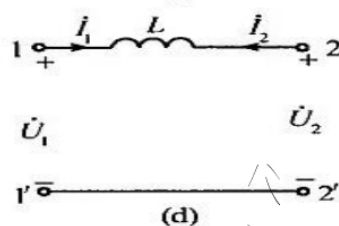
$$T = \begin{bmatrix} -1 & 0 \\ 1 & -1 \end{bmatrix}$$



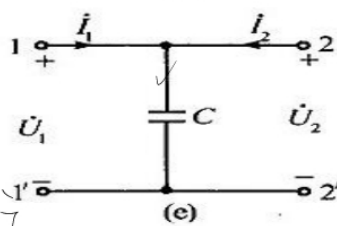
(a)



(b)

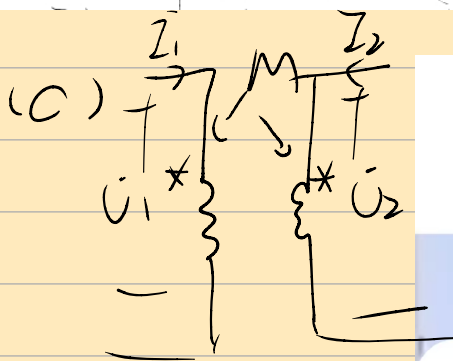


(c)



(d)

$$\begin{cases} \dot{U}_1 = \dot{U}_2 \\ \dot{I}_1 + \dot{I}_2 = j\omega C \dot{U}_2 \end{cases}$$



4:12

HD 5G 4G 72



结果1

2

3

全部解题完成 查看

解:(1) 题 16-3 图(a)所示电路的端口电压 $\dot{U}_1$ 、 $\dot{U}_2$ 和电流 $\dot{I}_1$ 、 $\dot{I}_2$ 的参考方向已经标出, T 参数方程为

$$\dot{U}_1 = \dot{U}_2, \quad \dot{I}_1 = -\dot{I}_2$$

所以 T 参数矩阵为

$$T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

(2) 按题 16-3 图(b)所标出的端口电压 $\dot{U}_1$ 、 $\dot{U}_2$ 和电流 $\dot{I}_1$ 、 $\dot{I}_2$ 的参考方向, 题 16-3 图(b)所示电路的 T 参数方程为

$$\dot{U}_1 = -\dot{U}_2, \quad \dot{I}_1 = \dot{I}_2$$

故 T 参数矩阵为

$$T = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

(3) 题 16-3 图(c)所示电路为耦合电感, 按已标出的端口电压 $\dot{U}_1$ 、 $\dot{U}_2$ 和电流 $\dot{I}_1$ 、 $\dot{I}_2$ 的参考方向, 其端口电压、电流满足方程

$$\dot{U}_1 = j\omega L_1 \dot{I}_1 + j\omega M \dot{I}_2 \quad (1)$$

$$\dot{U}_2 = j\omega M \dot{I}_1 + j\omega L_2 \dot{I}_2 \quad (2)$$

由式②, 得

$$\dot{I}_1 = \frac{1}{j\omega M} \dot{U}_2 - \frac{L_2}{M} \dot{I}_2 \quad (3)$$

将式③代入到式①中, 可得

$$\dot{U}_1 = \frac{L_1}{M} \dot{U}_2 - j\omega \frac{L_1 L_2 - M^2}{M} \dot{I}_2 \quad (4)$$

式④与式③联立可得 T 参数矩阵为

$$T = \begin{bmatrix} \frac{L_1}{M} & j\omega \frac{L_1 L_2 - M^2}{M} \\ -j\frac{1}{\omega M} & \frac{L_2}{M} \end{bmatrix}$$

(4) 对题 16-3 图(d)所示电路,按标出的端口电压和电流的参考方向,其 T 参数方程为

$$\dot{U}_1 = \dot{U}_2 - j\omega L \dot{I}_2, \dot{I}_1 = -\dot{I}_2$$

故 T 参数矩阵为

$$T = \begin{bmatrix} 1 & j\omega L \\ 0 & 1 \end{bmatrix}$$

(5) 对题 16-3 图(e)所示电路,按标出的端口电压和电流的参考方向,其 T 参数方程为

$$\dot{U}_1 = \dot{U}_2, \dot{I}_1 = j\omega C \dot{U}_2 - \dot{I}_2$$

故 T 参数矩阵为

$$T = \begin{bmatrix} 1 & 0 \\ j\omega C & 1 \end{bmatrix}$$

本题有不懂,随时问我



再拍一题

